

QQPlots and Normality Testing

MA2003B Application of Multivariate Methods
in Data Science

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However, many real-world datasets do not follow a normal distribution.

In this presentation, we will explore some other distributions and show how the concepts of skewness and kurtosis can help us understand its shape.

We will also introduce QQ plots, which are useful graphical tools for comparing the distribution of a dataset against a theoretical distribution, which is usually taken to be normal.

Skewness

Definition

The **skewness** of a random variable X is defined as the third standardized moment, given by

$$\gamma_1 = \frac{\mu_3}{\sigma^3} = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3},$$

where $\mu = \mathbb{E}[X]$ is the mean and $\sigma = \sqrt{\mathbb{E}[(X - \mu)^2]}$ is the standard deviation of X .

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Skewness measures the asymmetry of a probability distribution.

A distribution is said to be **positively skewed** (or right-skewed) if it has a longer tail on the right side, and **negatively skewed** (or left-skewed) if it has a longer tail on the left side.

Gamma Distribution

Definition

We say that a continuous random variable X follows a **Gamma distribution** with shape parameter $\alpha > 0$ and rate parameter $\beta > 0$, if its associated probability density function (PDF) is given by

$$f_X(x; \alpha, \beta) = \begin{cases} \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, & x > 0, \\ 0, & x \leq 0, \end{cases} \quad (1)$$

where $\Gamma(\alpha)$ denotes the Gamma function, defined as

$$\Gamma(\alpha) = \int_0^\infty t^{\alpha-1} e^{-t} dt.$$

The parameters α and β control the shape and rate of the distribution, respectively.

Among other applications, the Gamma distribution is frequently used to model the number of total insurance claims over a given time period.

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Remark

The skewness of a Gamma distribution is given by

$$\gamma_1 = \frac{2}{\sqrt{\alpha}}.$$

This means that the skewness is always positive, and it approaches zero as α increases. In other words, the distribution becomes more symmetric as α becomes larger.

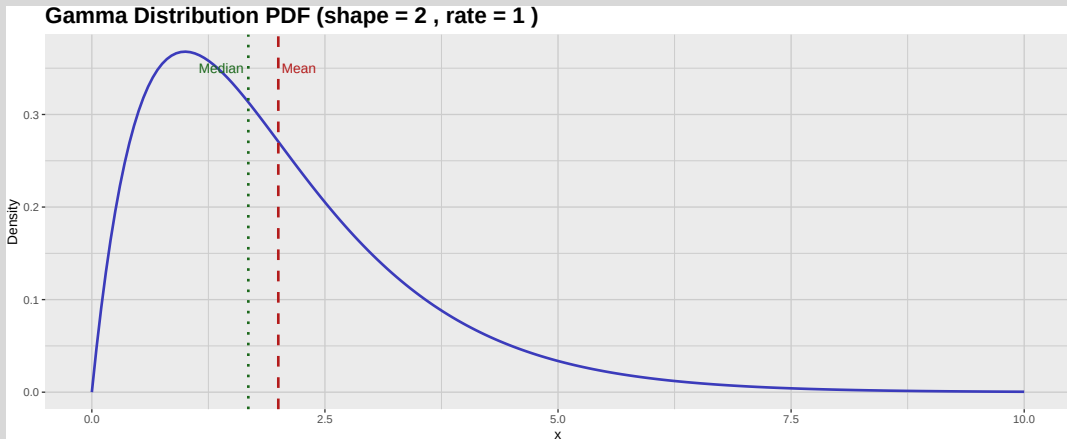


Figure 1: PDF of the gamma distribution with mean and median highlighted.

Kurtosis

Definition

The **kurtosis** of a random variable X is defined as the fourth standardized moment, given by

$$\beta_2 = \frac{\mu_4}{\sigma^4} = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4},$$

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The kurtosis of a normal distribution is 3, which is why frequently we are interested in computing the **excess kurtosis** which is given by:

$$\gamma_2 = \beta_2 - 3.$$

Laplace Distribution

Definition

We say that a continuous random variable X follows a **Laplace distribution** with location parameter $\mu \in \mathbb{R}$ and scale parameter $b > 0$, if its associated probability density function (PDF) is given by

$$f_X(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right), \quad x \in \mathbb{R}. \quad (2)$$

The parameters μ and b control the location and scale of the distribution, respectively.

Among other applications, the Laplace distribution is often used in image processing and computer vision, particularly in modeling noise in images.

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Remark

The excess kurtosis of a Laplace distribution is given by

$$\gamma_2 = 3.$$

This means that the Laplace distribution has heavier tails than a normal distribution.

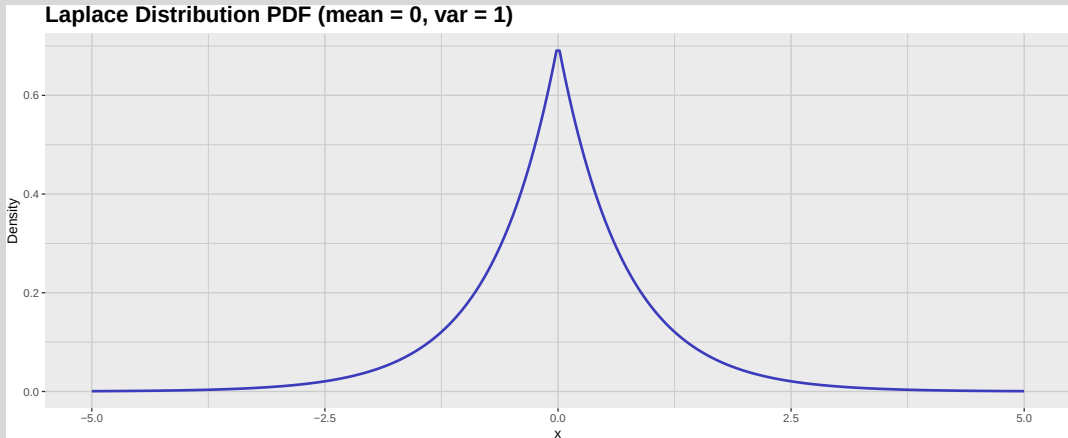


Figure 2: PDF of the Laplace distribution with mean zero and variance one.

QQ Plots

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They are particularly useful for assessing the normality of the data.

Constructing a QQ Plot

To construct a QQ plot, you should follow these steps:

1. **Sort the sample data:** Arrange the data in increasing order:

$$x_{(1)} \leq x_{(2)} \leq \cdots \leq x_{(n)}.$$

2. **Calculate plotting positions (empirical quantiles):** For each ordered value $x_{(i)}$, compute the corresponding probability

$$p_i = \frac{2i-1}{2n}, \quad i = 1, 2, \dots, n.$$

Constructing a QQ Plot

3. **Find theoretical quantiles of the normal distribution:** Compute the theoretical quantiles from the inverse cumulative distribution function (CDF) of the normal distribution:

$$q_i = \Phi^{-1}(p_i),$$

where Φ^{-1} is the quantile function of the standard normal distribution.

4. **Plot the points:** Plot the pairs $(q_i, x_{(i)})$ with q_i on the x-axis and $x_{(i)}$ on the y-axis.
5. **Add a reference line:** Typically, we add a line through the first and third quartiles to help assess normality.

The closer the points lie to this line, the more the sample resembles the specified normal distribution.

Gamma Distribution: QQ Plot and Histogram

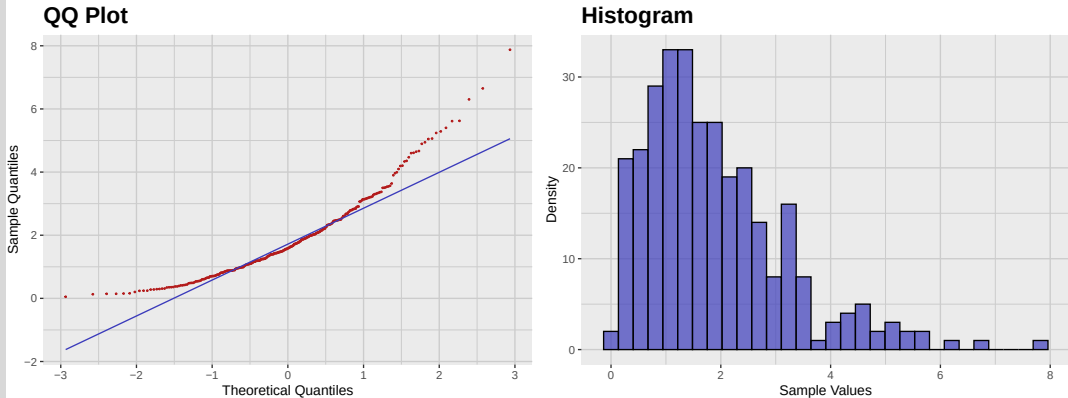


Figure 3: QQ plot and histogram of the Gamma distribution samples.

Laplace Distribution: QQ Plot and Histogram

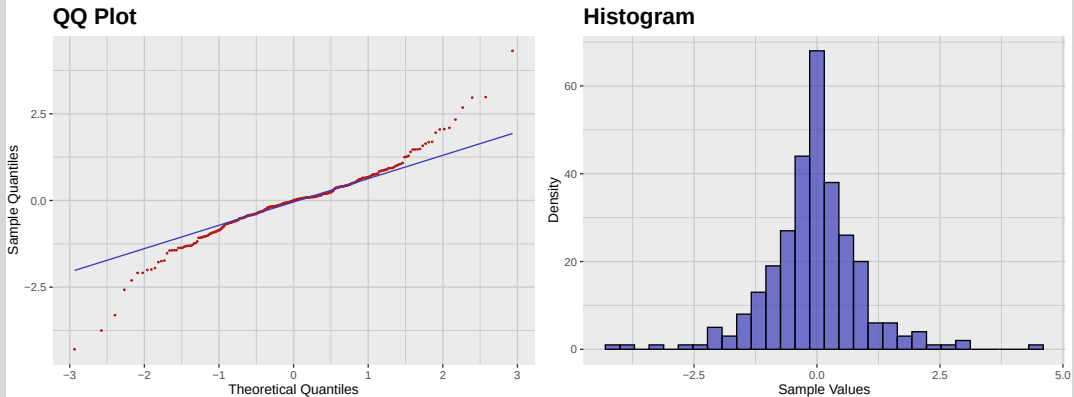


Figure 4: QQ plot and histogram of the Laplace distribution samples.

Sample Skewness and Kurtosis

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However, in practice, we often work with samples rather than entire populations.

Therefore, we need to define sample skewness and sample kurtosis.

Definition

Let $x_1, x_2, \dots, x_n$ be a sample of size n . The **sample mean** $\bar{x}$ and **sample variance** s^2 are given by

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2.$$

The **sample skewness** g_1 given by

$$g_1 = \frac{n}{(n-1)(n-2)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^3.$$

The **sample excess kurtosis** g_2 is given by

$$g_2 = \frac{n(n+1)}{(n-1)(n-2)(n-3)} \sum_{i=1}^n \left(\frac{x_i - \bar{x}}{s} \right)^4 - \frac{3(n-1)^2}{(n-2)(n-3)}.$$

As an example, we will compute the sample skewness and sample excess kurtosis of the Gamma and Laplace samples we showed earlier.

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For the Gamma sample we have:

```
Rows: 1
```

```
Columns: 2
```

```
$ sample_skewness      <dbl> 1.293772
```

```
$ sample_excess_kurtosis <dbl> -0.8356471
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```
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```

For the Laplace sample we have:

```
Rows: 1
Columns: 2
$ sample_skewness      <dbl> -0.07876743
$ sample_excess_kurtosis <dbl> 3.726374
```

Scaling and Standardization

In many situations, it could happen that the units we are using to measure different variables in our data are not compatible.

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In such cases, it is often useful to **scale** the variables so that they are comparable.

Scaling Random Variables

Definition

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$$Y = aX + b$$

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Remark

In the context of statistics, affine transformations are often referred to as **scaling** (even when they also include a translational component).

Now assume that X is a random variable with mean μ and standard deviation σ .

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This particular affine transformation is called a **standardization**, and the resulting random variable Z is called the **standardized** version of X . Z is also sometimes called the **z-score** of X .

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Remark

Notice that if X is not normally distributed, then Z will **not** be the standard normal distribution. Standardization does not make a non-normal distribution normal.

Observation

Notice that, by definition,

$$\gamma_1(Z) = \mathbb{E}[Z^3] = \frac{\mathbb{E}[(X - \mu)^3]}{\sigma^3} = \gamma_1(X)$$

and

$$\gamma_2(Z) = \mathbb{E}[Z^4] - 3 = \frac{\mathbb{E}[(X - \mu)^4]}{\sigma^4} - 3 = \gamma_2(X).$$

As an example, we will consider again the Gamma distribution that we introduced earlier. In this case we have that

$$\mu = \frac{\alpha}{\beta} \quad \text{and} \quad \sigma = \frac{\sqrt{\alpha}}{\beta}.$$

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Using this information, we can generate the plot of standardized version of the Gamma distribution and compare it with the original one.

Gamma Distribution: Original and Standardized

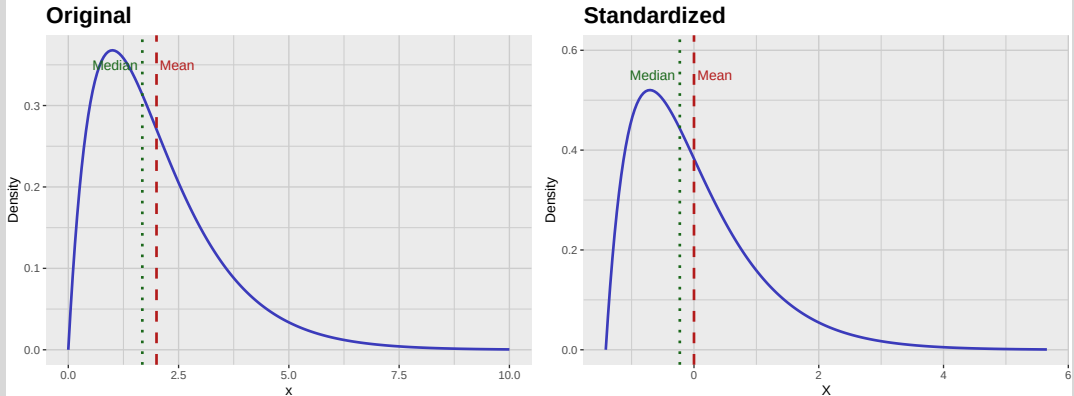


Figure 5: Comparison of the probability density function (PDF) of the original and standardized Gamma distribution.

Scaling Samples

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However, we can still try to scale our data but we should keep in mind that in this case there is more than one scaling method.

The most common are:

1. **Standardization:** This method uses the sample mean and sample standard deviation to scale the data. The standardized sample is given by

$$T = \frac{X - \bar{X}}{S},$$

where $\bar{X}$ is the sample mean and S is the sample standard deviation.

2. **Min-Max Scaling:** This method scales the data to a fixed range, usually $[0, 1]$. The scaled sample is given by

$$T = \frac{X - \min(X)}{\max(X) - \min(X)},$$

where $\min(X)$ and $\max(X)$ are the minimum and maximum values of the sample, respectively. This method is very sensitive to outliers, as it uses the minimum and maximum values of the sample.

3. **Robust Scaling:** This method scales the data using the median and interquartile range (IQR). The scaled sample is given by

$$T = \frac{X - \text{median}(X)}{\text{IQR}(X)},$$

where $\text{IQR}(X) = Q_3 - Q_1$ is the interquartile range of the sample. This method is more robust to outliers than min-max scaling, as it uses the median and IQR instead of the minimum and maximum values.

4. **Normalization:** This method simply divides each data point by the corresponding vector norm. The normalized sample is given by

$$T = \frac{X}{\|X\|},$$

where $\|X\|$ is the vector norm of the sample. It is not used as often as the previous methods.

Remark

As we mentioned earlier, scaling does not make a non-normal distribution normal. In particular, the sample skewness and sample excess kurtosis of the scaled sample will be the same as those of the original sample.

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As an example, we will consider the different scaling methods applied to the Gamma sample we generated earlier.

Comparison of Scaling Methods Applied to the Gamma Sample

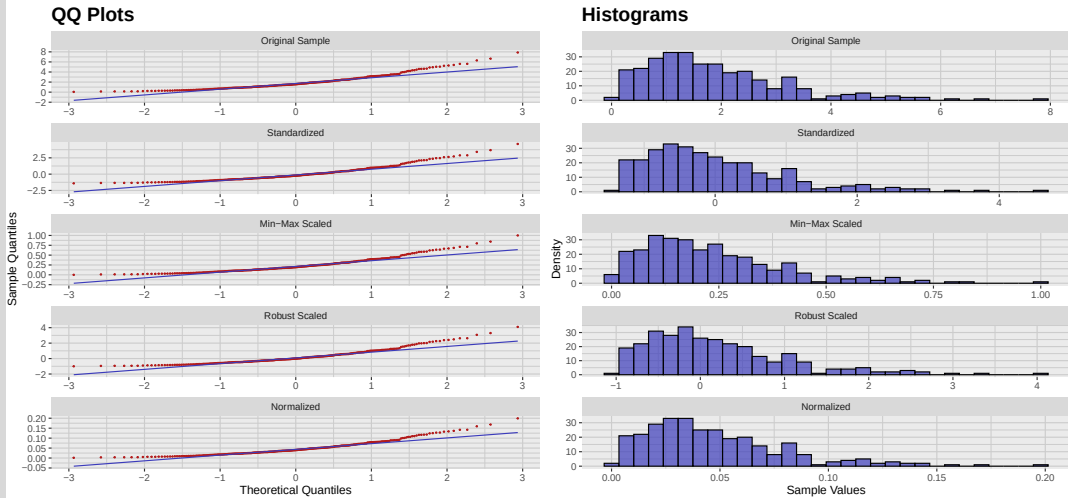


Figure 6: Comparison of different scaling methods applied to the Gamma distribution sample.

Activity: Visualizing the Non-Normality of Samples

For each of the following distributions, construct a sample of size 300 and compute their associated sample Skewness and Kurtosis. After that, construct and compare their associated QQ plots and histograms.

1. Exponential distribution with rate parameter $\lambda = 1$.
2. Uniform distribution on the interval $[0, 1]$.
3. Beta distribution with shape parameters $\alpha = 2$ and $\beta = 5$.
4. Cauchy distribution with location parameter $x_0 = 0$ and scale parameter $\gamma = 1$.
5. Chi-squared distribution with degrees of freedom $k = 3$.

Normality Testing

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Therefore, it becomes necessary to develop more objective and precise statistical tests to determine whether the data we are analyzing follows a normal distribution.

We will now explore some commonly used statistical tests for normality and discuss some transformations that can be performed on our data to make it more normal-like.

Normality Tests

There are several statistical tests that can be used to determine whether a dataset follows a normal distribution. Some of the most commonly used tests include:

1. **Shapiro-Wilk test:** This test is based on the correlation between the data and the corresponding normal scores. It is a powerful test for small sample sizes (less than 50) and is widely used in practice. However, it can be overly sensitive to small deviations from normality in larger sample sizes.
2. **Kolmogorov-Smirnov test:** This test compares the empirical distribution function of the data with the cumulative distribution function of the normal distribution. It is a non-parametric test and can be used for any sample size. However, it is less powerful than the Shapiro-Wilk test for small sample sizes. It is also sensitive to scale and location parameters, which can affect the results of the test.

3. **Anderson-Darling test:** This test is similar to the Kolmogorov-Smirnov test but gives more weight to the tails of the distribution. It is a powerful test for small sample sizes and is widely used in practice.
4. **Jarque-Bera test:** This test is based on the skewness and kurtosis of the data. It is a powerful test for large sample sizes and is widely used in practice. However, it can be overly sensitive to small deviations from normality in smaller sample sizes.

We will now apply these tests to the gamma sample data we created in the previous chapter:

The results of the normality tests are as follows:

```
# A tibble: 4 x 2
  test_factor      p_value
  <fct>          <dbl>
1 Shapiro-Wilk    1.58e-12
2 Kolmogorov-Smirnov 7.93e- 3
3 Anderson-Darling 9.02e-16
4 Jarque-Bera      0
```

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4 Jarque-Bera      0
```

As we can see, none of these tests was particularly successful, which is not surprising given that the gamma distribution is not normal.

Transformations

To make our data more normal-like, we can apply various transformations to any given sample. Some of the most common transformations include:

1. **Box-Cox transformation:** This is a family of power transformations that can be applied only to data that is always positive. It is given by the formula:

$$\tilde{X} = \begin{cases} \frac{X^\lambda - 1}{\lambda} & \text{if } \lambda \neq 0 \\ \log(X) & \text{if } \lambda = 0 \end{cases}$$

where λ is some parameter.

2. **Yeo-Johnson transformation:** This transformation is similar to the Box-Cox transformation but can handle negative values. It is useful for data that is positively or negatively skewed. It is given by the formula:

$$\tilde{X} = \begin{cases} \frac{((X+1)^\lambda - 1)}{\lambda} & \text{if } X \geq 0, \lambda \neq 0 \\ \frac{-((-X+1)^{2-\lambda} - 1)}{2-\lambda} & \text{if } X < 0, \lambda \neq 2 \\ \log(X+1) & \text{if } X \geq 0, \lambda = 0 \\ \log(-X+1) & \text{if } X < 0, \lambda = 2 \end{cases}$$

where λ is some parameter.

We will now apply the Box-Cox transformation to the gamma sample data with parameters $\lambda = 0, 0.25, 0.5, 0.75$, and 1:

Comparison of Box-Cox Transformations with Different Lambda Values

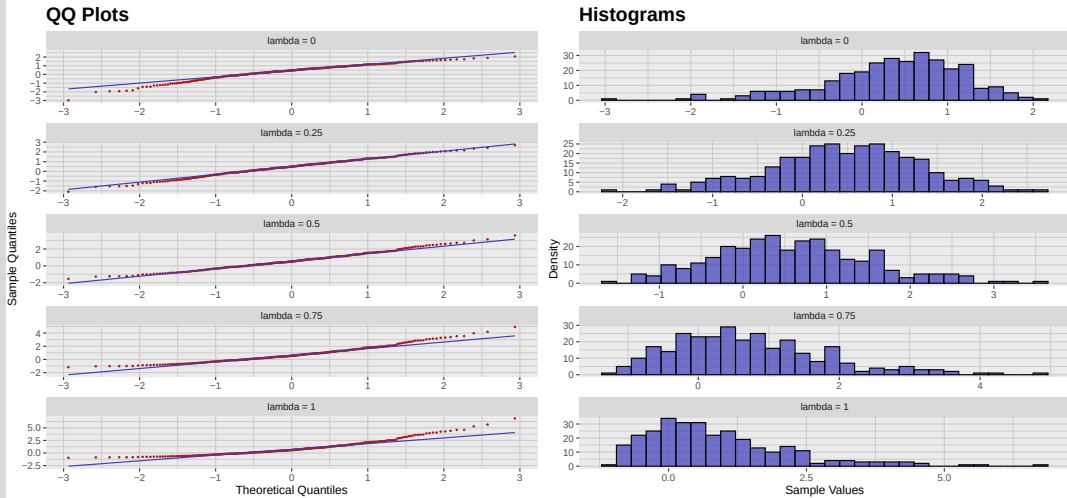


Figure 7: Comparison of different Box-Cox transformations.

Normality Score Optimization

We wish to find the optimal value of λ that maximizes the log-likelihood of the transformed data.

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The following figure shows the log-likelihood profile for the Box-Cox transformation applied to the gamma sample data:

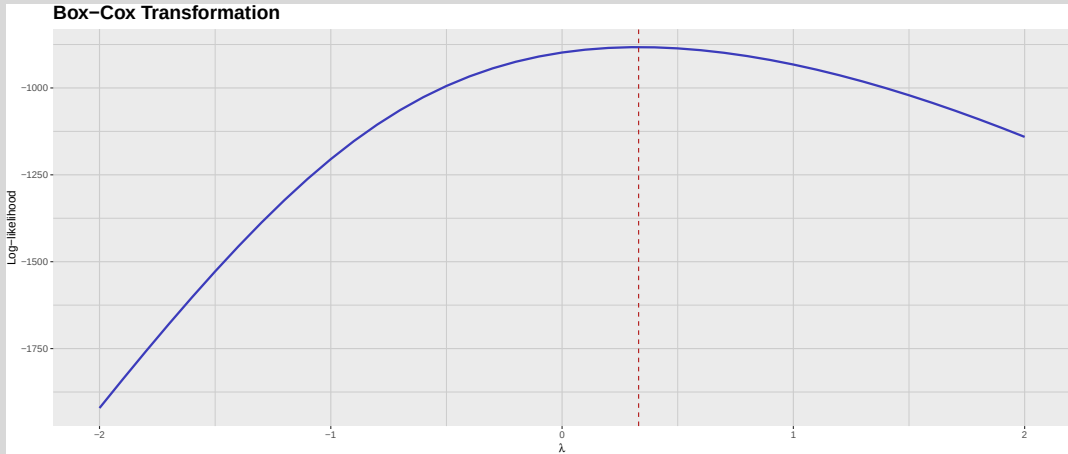


Figure 8: Log-likelihood profile for the Box-Cox transformation.

Finally, we can extract the optimal value of λ and transform the data using this value:

```
# A tibble: 4 x 2
  test_factor      p_value
  <fct>          <dbl>
1 Shapiro-Wilk    0.974
2 Kolmogorov-Smirnov 1.000
3 Anderson-Darling 0.977
4 Jarque-Bera     0.878
```

Finally, we can extract the optimal value of λ and transform the data using this value:

```
# A tibble: 4 x 2
  test_factor      p_value
  <fct>          <dbl>
1 Shapiro-Wilk    0.974
2 Kolmogorov-Smirnov 1.000
3 Anderson-Darling 0.977
4 Jarque-Bera     0.878
```

As we can see, the Box-Cox transformation with the optimal value of λ significantly improves the normality of the data, as all the normality tests now yield high p -values. We now show the QQ plot and histogram of the transformed data:

Optimal Gamma Distribution Transformation

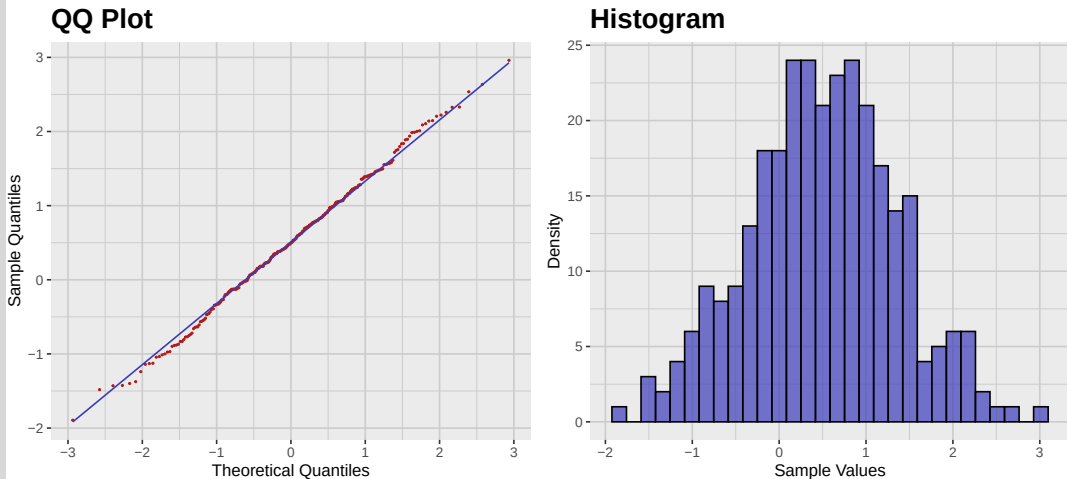


Figure 9: QQ plot and histogram of the Optimal Gamma Distribution Transformation.